<table>
<tr><td colspan="3">National Institute of Business Management
School of Computing and Engineering
Course work | Assessment Announcement Sheet</td><td>  </td></tr>
<tr><td colspan="2">Course Name</td><td colspan="2">HNDSE</td></tr>
<tr><td colspan="2">Module Name</td><td colspan="2">Internet of Things</td></tr>
<tr><td colspan="2">Batch</td><td colspan="2">26.2F</td></tr>
<tr><td colspan="2">Learning Outcomes Covered (Mention according to the Module Descriptor)</td><td colspan="2">1. Explain the concept of IoT with their building blocks of IOT.
2. Apply communication protocols and edge computing.
3. Ability to work with lead IOT platforms.
4. Apply machine learning models to IOT applications.
5. Ability to explain the future trends in IOT with the implement power management techniques.</td></tr>
<tr><td colspan="2">Assessment | CW No</td><td colspan="2">Coursework 1</td></tr>
<tr><td rowspan="3">Assessment Mode</td><td rowspan="3">Group</td><td colspan="2">Group (if it is group mode only)</td></tr>
<tr><td>Group Size</td><td>Grouping Criteria</td></tr>
<tr><td>3</td><td>Grouped by Students</td></tr>
<tr><td rowspan="2">Assessment Type</td><td colspan="3"></td></tr>
<tr><td>If other specify</td><td colspan="2"></td></tr>
<tr><td colspan="2">Hand in Date | Time</td><td colspan="2">02/09/2026</td></tr>
<tr><td colspan="2">Hand out Date | Time</td><td colspan="2">02/10/2026</td></tr>
<tr><td colspan="2">Submission Details (Format and Location)</td><td colspan="2">Presentation / Demonstration with Viva/Report</td></tr>
<tr><td colspan="2">Plagiarism Criteria</td><td colspan="2">None</td></tr>
</table>

Assessment | CW Description

Introduction

Internet of things or IoT is a system of connected devices through the internet. It involves mechanical devices, sensors, home appliances, vehicles, etc., apart from desktop, mobile, and laptop. These devices are designed in such a way that they can share data with other devices over the internet. IoT basically provides a platform for devices to interact and collaborate with each other.

Scenario

In this course work you are assigned to do a research project to create an IOT application which is useful in our day today life using Nodemcu or Raspberry pi or Arduino R4 or any micro controller with WIFI.

Grading Criteria:

- Proposal Defense. (10 Marks)
- 1st Progress. (10 Marks)
- 2nd Progress (10 Marks)
- Final Presentation / Demonstration with Viva/Report (70 Marks)

Proposal Defense: (ILO 1,2)

- Prepare a maximum of three pages Synopsis for your proposed research project.
- **Synopsis** is the plan for your research project. It provides the rationale for the research, the research objectives, expected functionalities and the proposed methods.
- **Proposal Should include followings.**
  - Introduction to Proposed Idea
  - Block Diagram/Rough Design
  - Budget
  - Time Frame
- Submit the PDF of the proposal to LMS.

1st Progress: (ILO 2, 3 ,4) Date:18th September - 2026

- Students must bring and submit the following documents to the 1st progress meeting.

1st Progress	Literature Reviews Detailed Design (Sketch or 3D design) Completed mobile Application and Web Application Demonstration of Partially Completed Prototype

2nd Progress: (ILO 2, 3 ,5) Date: 18th September - 2026

- Students must submit the following document to the LMS.

2nd Progress	1st Draft of the Report Fully Completed Prototype

Report should have

- Abstract
- Acknowledgment
- Table of Content
- Table of Figures
- Introduction
- Methodology
- Discussion
- Future Implementations
- References
- Gantt Chart

Final Presentation / Demonstration with Viva/Report Submission: (ILO 1, 2 ,3, 4, 5) s

1. Completed report Should be submitted to the LMS. (Similarity< 15%, AI < 5%)
   Date of Submission – 02nd **October-2026**

2. Extended Abstract should be submitted to the LMS.
   Date of Submission – 02nd **October-2026**

3. Final Presentation and VIVA

s

Criteria	70%+	60%-69%	50%-59%	40%-49%	0%-39%
Proposal Defense (10 Marks)	Clear identification and definition of problem that can be automated. Welldeveloped solution through the six steps provided. Strong rationale for provided idea.	Identification and definition of problem is mostly clear, with some minor gaps. Solution adequately illustrates relationships but lacks depth. Adequate rationale for ideas but may be inconsistent or underdeveloped.	Basic identification and definition of problem, with noticeable gaps. Solution outlines relationships but lacks coherence or clarity. Limited rationale for ideas, with some inaccuracies.	Identification of problem is incomplete or unclear. Solution lack's structure or relevance. Weak rationale or justification for articulated ideas.	No attempt made at providing a solution. Complete lack of understanding or relevance of ideas. No rationale provided.
1st Progress (10 Marks)	Structured all the documentation clearly and relevant. Documentation formats are followed exceptionally.	Structured all the documentation clearly and relevant enough. Documentation formats are followed.	Structured all the documentation lacks clearly and relevant. Documentation formats are not followed.	Not structured all the documentation clearly and relevant. Documentation formats are not followed .	No documentations provided highlighting understanding or with relevance. No formatting structure used.
2nd Progress (10 Marks)	Structured all the documentation clearly and relevant. Documentation formats are followed exceptionally. Mechanical system exceptionally implemented.	Structured all the documentation clearly and relevant enough. Documentation formats are followed. Mechanical system implemented correctly.	Structured all the documentation lacks clearly and relevant. Documentation formats are not followed. Mechanical systems need improvement.	Not structured all the documentation clearly and relevant. Documentation formats are not followed. The mechanical system is not complete.	No documentations provided highlighting understanding or with relevance. No formatting structure used. No mechanical system.

Final Presentation (10 Marks)	Delivery of a concise and engaging presentation within time limits. Clear articulation of key findings and implications.	The presentation is mostly concise and engaging. Key findings are articulated clearly. Responses to questions are effective.	Presentation lacks conciseness or engagement. Key findings may lack clarity. Responses to questions are adequate.	The presentation is unclear or lacks engagement. Key findings are poorly articulated. Responses to questions are insufficient or inaccurate.	No presentation provided. Lack of understanding in presenting findings. Ineffective responses to questions. Poor academic
	Effective responses to questions.	Very good academic presentation skills demonstrated.	Good academic presentation skills shown.	Reasonable academic presentation skills displayed.	presentation skills evident.
Demonstration (35 Marks)	Quick and clear demonstration of all the functions in the developed robot. Reflect exceptional preparation that goes behind demonstration.	Clear demonstration of some of the functions working in the developed robot. Reflect preparation that goes behind demonstration.	Demonstration is somewhat clear. Most functions are working. Evidence for some nonfunctional functions is provided.	Demonstration is not clear. Most functions are not working. Evidence to most nonfunctional functions isn't provided.	No demonstration. Function are not working.
Viva (25 Marks)	Quick and well-organized answers demonstrating the understanding and highlighting the contribution to the project.	Mostly correct answers demonstrating most understanding highlighting contribution to the project.	Answers lack understanding and does not highlight required contribution to the project.	Lack clear nor correct answers to the questions asked. Does not reflect required contribution to the project.	No clear nor correct answers provided. No contribution identified.